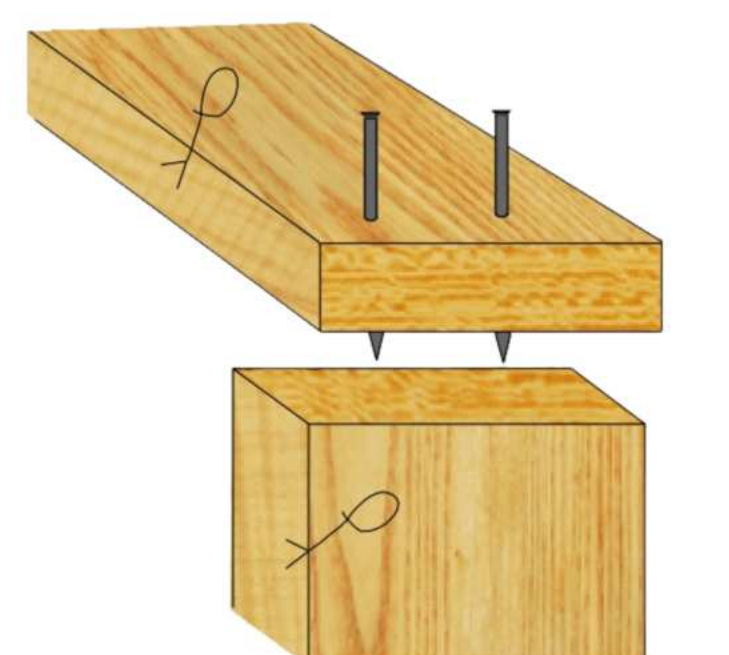


REDUCING NAIL WASTE: EVALUATING THE APPLICABILITY OF TREATED WOOD AND POLYMER IN FASTENER FREE JOINTS

Introduction

- Modern Furniture mainly uses joints with metal fasteners.
 - End-to-End joint commonly used due to simplicity and compressive tolerance (College of Fine and Applied Arts, 2014).
 - Nails result in millions of tons of Metal-Nails waste (Attia et.al, 2022).
- Sashimono
 - Japanese wood-joinery (Pontsioen, 2012)
 - Mortise Tenon joint well known for compressive strength tolerance (Camacho etL.al, 2018)
 - Fastener-Free joint
- 3D Printing prevalent and prominent in construction (Camacho, 2018)



(Woodworking Stack Exchange, 2014)



(Shopify, 2022)

Problem Variables and Hypothesis

Research Question: Determining the extent to which the compressive strength of fastener free joints compares with that of fastener joints.

IV: Mortise Tenon Wood Joint & hard Resin Mortise Tenon Joint.

DV: Compressive Strength in Newtons of the joint.

Control: End to End Wood Joint & End to End Hard Resin Joint

Hypothesis: Untreated Pinewood mortise-tenon joints will demonstrate the greatest mechanical compressive strength followed by hard resin mortise tenon joints.

Methodology



Materials

- Untreated Southern Pine Wood
- V1 1500 Low Warp Resin
- Formlabs 3D Printer
- Formlabs Form Cure
- Model H7789 Mortising Kit
- Drill Press
- Belt Saw
- Belt Sander
- Mark 10 Force Stand

Conclusions

We found that there are no statistically significant differences between the compressive strength of wood fastener free joints and wood fastener joints. Moreover, we found there to be no significant difference between hard resin fastener free and fastener joints. It is important the high likelihood of type 1 error for all the groups. This error implies there to be an incorrect assumption about the null hypothesis (False, Positive). Other noteworthy error is how we defined mechanical failure. We attempted to print the resin joints with infill densities similar to the density of wood but were not able to do so exactly. There is also uncertainty because the force sensor could not detect compressive strength above 1500 newtons. As a result of these pieces of error we cannot definitively state that there is or there is not a difference between fastener and free joints and the materials used in terms of compressive strength.

Results

We conducted a mann whitney u test for our data since we did not have more than 30 trials and did not know if our data was normally distributed. The p-value, significance value, was 0.05 for all tests conducted.

- We failed to reject the null hypothesis for the resin end to end and resin mortise tenon joints group.
- We also failed to reject the null hypothesis for the wood mortise tenon vs the wood end to end joints.
- There was also no statistically significant difference between the hard resin and wood end to end joints.
- There was alos no statistically significant difference between the wood and resin mortise tenon joints.

Tough V1 Low-Warp 1500 Resin Mortise Tenon Joint Test

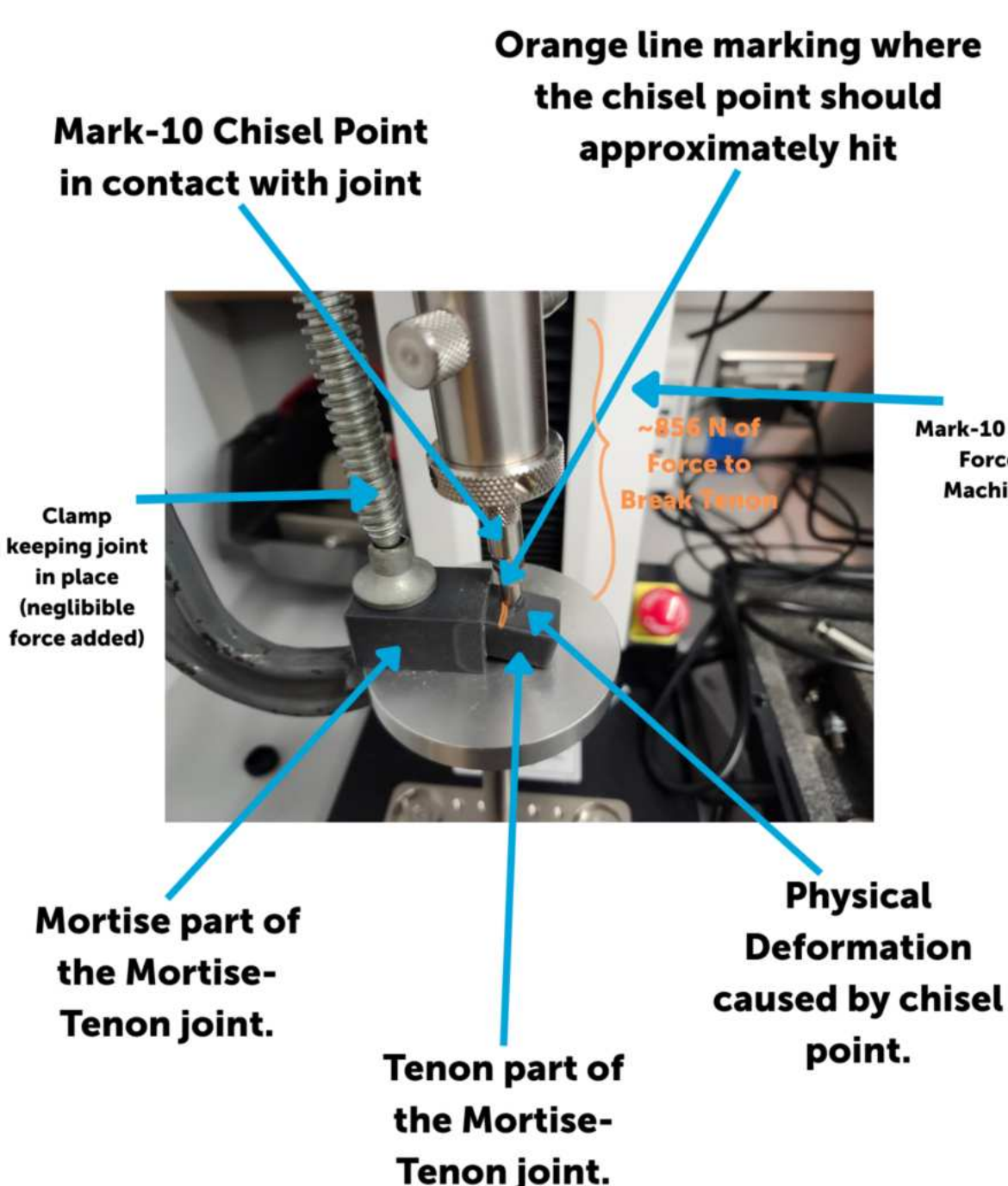
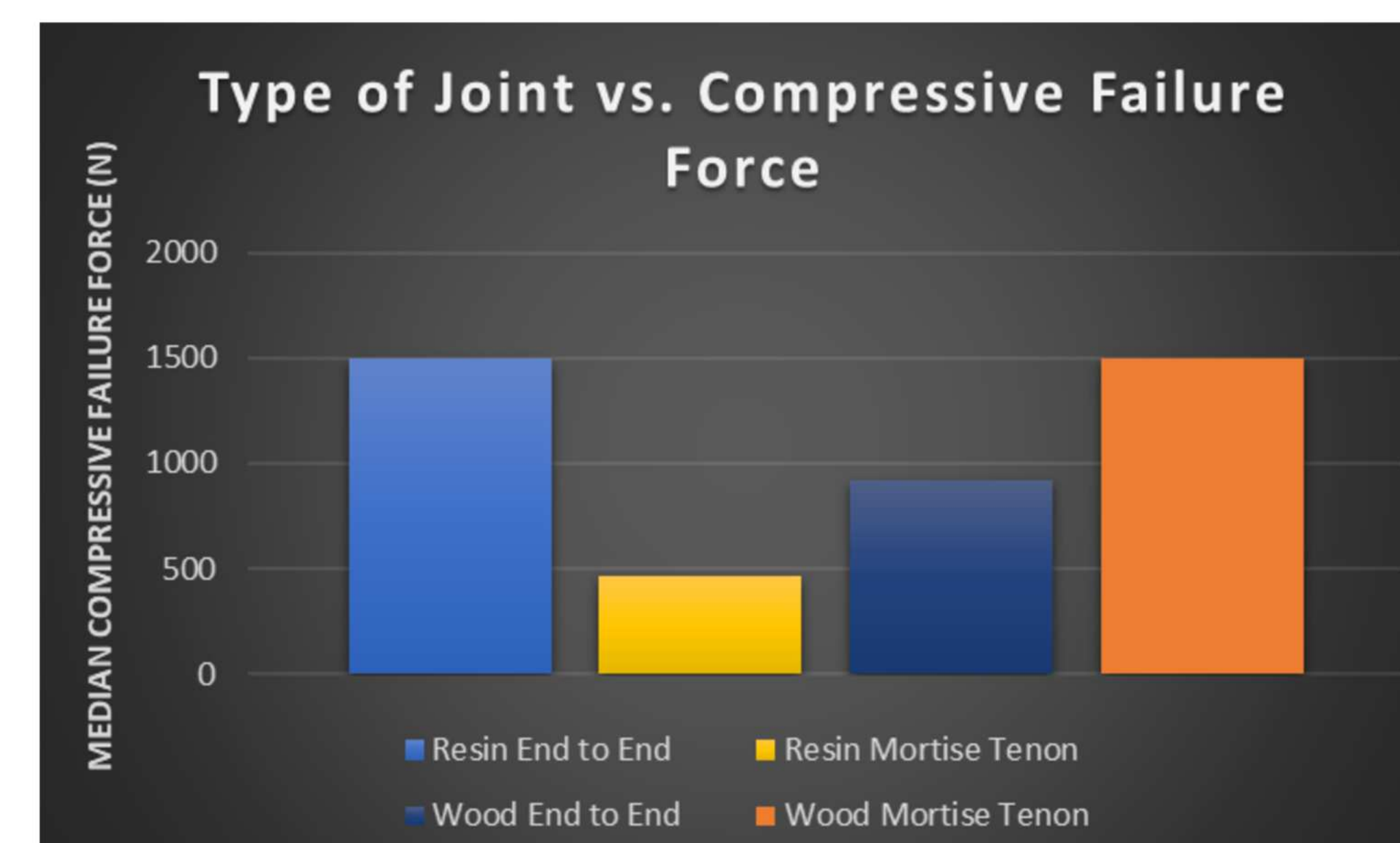


Table 1

System Information:				
IntelIMESUR Version: 2.0.2.1				
Test Frame Model: F305				
Test Frame Serial Number: 4338572				
Test Frame Capacity: 300 lbf / 1.5 kN				
Test Frame Firmware Version: 1.00.14				
Force Sensor Model: FS05-300				
Force Sensor Capacity: 1500 N				
Force Sensor Resolution: 0.5 N				
Force Sensor Serial Number: 4339048				
Calibration Date: May 31, 2023				
Calibration Due Date: May 31, 2024				
Reading	Load [N]	Distance [mm]	Time [s]	
567	1316.5	-22.78	5.66	
568	1315.5	-22.85	5.67	
566	1314	-22.76	5.65	
569	1312	-22.89	5.68	
565	1310	-22.72	5.64	
570	1309	-22.93	5.69	
571	1307.5	-22.98	5.7	
572	1306.5	-23.02	5.71	

Note: Table above shows data collected from a wood mortise tenon trial

Graph 1



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